25900749

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Cre Course (DSC) - MG1DSCBTG100

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

Part A

Answer the Questions in One Word or Sentence-Answer All Questions

Each Question carries 1 mark

1. Name the type of cell that synthesizes immunoglobulin. [U] / [CO3]
   Plasma cells/B cells-1 mark
2. Which enzyme unwinds the DNA helix at the replication fork? [U] / [CO2]
   Helicases- 1 mark
3. Give an example of conventional biotechnology. [K] / [CO1, CO2]
   Fermentation/ any one example- 1 mark.
4. Name two industries where biotechnology is applied. [K] / [CO1, CO2]
   Any two industries- 1 mark.
5. Who contributed to the development of cell theory alongside Mathias Jakob Schleiden. [K] / [CO1, CO2]
   Theodor Shwaan- 1mark
6. State the different types of nitrogen bases. [K] / [CO1, CO2]
   Purines and pyrimidines/4 bases- 1mark
7. Who is the father of rDNA technology [U] / [CO2]
   Paul Berg-1 mark
8. What is the main reproductive structure of fungi? [U] / [CO1, CO2]
   Sporangium/conidiophores or conidia-1 mark
9. What type of RNA carries genetic information? [K] / [CO2]
   mRNA- 1 mark
10. Which company was the first to commercialize Humulin? [K] / [CO1, CO2]
    Eli Lilly and company- 1 mark

, **Part B**

Short Answer Questions-Answer any 4 Questions

Each Question carries 3 marks

11. Give an outline of major milestones in biotechnology. [K] / [CO1, CO2]
    Major milestones-Any 3 points- 3 marks.
12. How does the secondary structure of a protein form? [K] / [CO1, CO2]
    Secondary structure a protein- introduction- 1 mark, explanation- 2 marks.
13. Explain monohybrid and dihybrid crosses with examples. [U] / [CO3]
    Monohybrid and dihybrid crosses- explain with examples- 3marks
14. Describe the solubility properties of carbohydrates in water and explain their significance. [U] / [CO1, CO2]
    Carbohydrates- introduction/solubility properties (4 points)- 2 marks, significance- 1mark
15. State the function of golgi apparatus? [K] / [CO1, CO2]
    Golgi apparatus- introduction- 1mark, functions- 2 marks.
16. Describe the components in the structure of a virus. [K] / [CO1, CO2]
    Virus- introduction- 1 mark, structure--components- 2 marks [4x3 = 12]

Part C

Short Essay Questions-Answer any 3 Questions

Each Question carries 6 marks

17. Summarize the structures and functions of prokaryotic and eukaryotic cells, highlighting the key differences.
    Structure- difference- explain- 3 marks, functions- 3 marks [U] / [CO1, CO2]
18. Explain the action of recombinant vaccines in COVID-19. [U] / [CO2]
    Recombinant vaccines- introduction-2 marks, COVID-19 vaccine- principle, explanation- 4 marks
19. What is the role of mRNA in translation? [U] / [CO2]
    mRNA- introduction, structure-2 marks, role of mRNA in translation- 4 marks
20. Explain the history of biotechnology? [U] / [CO1, CO2]
    History of biotechnology- Major milestones- (6 points)- 6 marks
21. State the three fundamental laws of inheritance proposed by Gregor Mendel. [K] / [CO3]
    Three Laws- explain-2 marks each- 6 marks [3x6 = 18]

Part D

Essay Questions-Answer any 1 Question

Each Question carries 10 marks

22. Outline the composition and functions of phospholipids in biological membranes. [K] / [CO1, CO2]
    Biological membranes- introduction-models- 3 marks, composition and functions-minimum 7 points- 7 marks
23. Determine the important cells involved in immune system. [An] / [CO3]
    Immune system- introduction- 2marks, cells involved- explain- 8 marks.

[1x10 = 10